

Contents

External Adversarial Audit – Paper III preprint draft v1.3	1
1. Overall verdict	1
FAIL	1
2. E0 – intake, provenance and sealing: PASS	2
3. E1 – claims and scope: PASS	2
4. E2 – independent rederivation: INCONCLUSIVE	3
4.1 What was rederived, exactly	3
4.2 What was not attempted, named individually	3
5. E3 – formal conformance: PASS	4
6. E4 – Lean reproduction and axiom boundary: FAIL	5
6.1 The clean build was achieved	5
6.2 Why the gate nonetheless fails	5
6.3 The author’s build record is a replay, not a build	5
7. E5 – bilingual, format and render: FAIL	6
7.1 What passes, and it is most of it	6
7.2 The divergence	6
8. E6 – citations, prior art, novelty: INCONCLUSIVE	7
9. E7 – release-package integrity: FAIL	7
10. Mandatory regression matrix	8
11. Findings ledger	8
12. What this audit does and does not establish	9
13. Signature	9

External Adversarial Audit – Paper III preprint draft v1.3

Request: EXTERNAL_ADVERSARIAL_AUDIT_REQUEST.md

Request SHA-256:

a0569fa455d70ee3e8af8ddb4362c52863de69d830bc76975bb6f9d22d9e12c0

Target: preprints/PAPER_III/active/preprint_draft_v1.3/ **Run:** run_2026-08-22_v1.3 **Dates:** 2026-08-22

Auditor: Claude (Anthropic), claude-opus-5. **No adversarial challenger.**

Only v1.3 was evaluated. No verdict, PASS, or conclusion from any earlier run was inherited.

1. Overall verdict

FAIL

Two gates fail and two are inconclusive. **No mathematical or formal defect was found**, and that distinction carries the whole weight of this report: the theorem compiles, its axiom footprint is clean, and every inequality this auditor rederived held exactly. The failures are in the **build configuration** and in **bilingual fidelity** – release-readiness defects in a release candidate.

Gate	Verdict	Basis
E0 intake, provenance, sealing	PASS	every declared hash and manifest verifies; archive structurally clean
E1 mathematical claims and scope	PASS	claims, quantifiers and scope separation all confirmed
E2 independent rederivation	INCONCLUSIVE	Sections 4 and 9 rederived exactly; K-EPS, K-CORRIDOR, K-SPARSE not attempted
E3 formal statement conformance	PASS	validated against an auditor-constructed bridge, not by compilation

Gate	Verdict	Basis
E4 Lean reproduction and axioms	FAIL	the public aggregate root does not import the canonical theorem path
E5 bilingual, format, render	FAIL	EN/ES divergence in the content of a numbered proposition
E6 citations, prior art, novelty	INCONCLUSIVE	no source retrieved in this run; no specialist search possible
E7 release-package integrity	FAIL	freeze metadata contradicts its own package

Findings: **0 blocking, 0 critical, 3 major, 4 minor.**

2. E0 – intake, provenance and sealing: PASS

Every integrity chain verifies against the bytes received.

Check	Result
04_integrity/CURRENT_TARGET_SHA256.txt	12/12 , LF-only
manuscript sidecar	6/6 , LF-only
freeze-archive sidecar	1/1 , LF-only
SOURCE_MANIFEST.sha256	707/707
PACKAGE_MANIFEST.sha256	742/742
freeze ZIP SHA-256 vs the value declared in the request	2eb0ff20...d300, match
CRC of all 743 archive members	intact

Archive hygiene, every category zero: path traversal, absolute paths, backslash names, symlink/reparse entries, compiled project artifacts (`.olean`, `.ilean`, `.o`, `.c`, ...), `.lake` directories, duplicate entries.

Auditor’s own manifest of the received target: **875 files, 16,502,156 bytes**, recorded at `00_CONTROL/TARGET_SHA256.txt`. `05_formalization/lean_v1.3_candidate/` is excluded, as the request directs.

Thirteen stale-version strings appear in the tree; twelve are legitimate provenance or baseline references, and the two in `FREEZE_REPORT.md` and `CHANGELOG_v1.3.md` are explicit statements of what v1.3 descends from. The publication artifacts themselves carry **no** stale label.

3. E1 – claims and scope: PASS

Theorem 1.1 and Corollary 1.2 assert, with an **existential** absolute constant:

exists C , for every split graph G on n vertices: $|E(G)| - 2 \nu_3(G) \leq n^2/6 + C n$
exists C , for every split graph G on n vertices: $cp(G) \leq n^2/6 + C n$

The paper claims exactly what the request says it should, and not more. It states that Corollary 1.2 “improves the quadratic coefficient from $3/16$ to the sharp value $1/6$ ”, that this matches the lower bound at quadratic order, and – three times in English and three times in Spanish – that the corresponding statement for general chordal graphs **remains open**. It separates the sharp quadratic coefficient from the “still-undetermined least uniform linear coefficient”, which is the two-sharpness-roles distinction the request asks about.

Zero occurrences, in either language, of wording claiming to resolve or settle the split case or Erdős Problem #81. The previously recorded overbroad-scope finding is closed.

4. E2 – independent rederivation: INCONCLUSIVE

4.1 What was rederived, exactly

Section 9, the proof of the main theorem, was rederived in exact symbolic algebra from the manuscript's own displayed inequalities. **14 of 15 items PASS, one PARTIAL, none fails.**

K-COVER is closed. The case split is exhaustive and non-overlapping:

- $q \geq 2p-1$ versus $q < 2p-1$ is complementary over the integers, and the second gives $\alpha = q/p < 2 - 1/p < 2$, so α lies in $[0, 2)$.
- Every sequence α_k in $[0, 2)$ has, by Bolzano-Weierstrass on the compact $[0, 2]$, a convergent subsequence. Interior limit gives the bulk regime with $\epsilon_p = \min(L, 2-L)/2 > 0$; limit 0 gives Section 9.2; limit 2 gives Section 9.3. The three are mutually exclusive and exhaust the possibilities, which is exactly what a contradiction-over-subsequences argument requires.
- Inside Section 9.3, $s = 0(\sqrt{p})$ and $s/\sqrt{p} \rightarrow \infty$ are complementary by construction, and $s = \Theta(p)$ cannot occur since $\alpha \rightarrow 2$ forces $s = o(p)$.
- The dispersion dichotomy $D \geq q s^2/12$ or $D < q s^2/12$ is trichotomy of the reals.

The numbered inequalities, each verified exactly:

Item	Result
(9.4) $p \geq 2304$	forced, not chosen: the window $6\sqrt{p} \leq s \leq p/8$ is non-empty exactly when $48 \leq \sqrt{p}$, i.e. $p \geq 2304$
(9.5) $3m \leq s-3$	valid: $k \geq 1$ gives $3m < s - 5/2$, and integrality rounds to $s-3$
(9.10) $\delta \geq 7/8$	both parities; the odd case $(p-s)/p$ is binding and attains equality exactly at $s = p/8$
(9.11) parabola bound	exact: the maximum over the admissible range is at $x=(s-3)/3$ and equals $2s(s-3)/9$ identically
(9.12) high-dispersion contradiction	exact: $5 \cdot 36/288 = 5/8$ and $1/2 - 5/8 = -1/8$, so the sign flip holds at the boundary and strictly beyond
$\theta_R \leq 8s/(23p)$	exact at the two extreme substitutions $\rho = s/3$, $s = p/8$
(9.16) $\kappa_R \leq 5s/(4p)$	holds, at about $1.229 s/p$ against the allowed $1.25 s/p$ – valid but not generous
(9.18)	reduces exactly to $s \leq p/8$, consuming the whole hypothesis budget
(9.19) completion of the square	exact identity , so the bound $s^2/24$ is attained at $\rho = s/4$ and loses nothing
(9.20) low-dispersion contradiction	exact: $36/64 = 9/16$ and $1/2 - 9/16 = -1/16$

Section 4 was rederived in the same way in the immediately preceding audit of the unchanged mathematics and is not re-reported here; Section 9 is the new work.

The one PARTIAL: the Lemma 7.1 hypothesis chain in the low-dispersion branch. The arithmetic checks exactly ($q \geq 15p/8$ from $s \leq p/8$; $b = p - \rho > p - s/3$; $b \geq 2$ with enormous room since $p \geq 2304$), but the chain $2\rho + t_i + 1 \leq 3m + 1 \leq s - 2$ needs the definitional facts $\rho \leq m$ and $t_i \leq m$ from Section 7, which the auditor did not rederive.

4.2 What was not attempted, named individually

- K-EPS: the AX1/nibble epsilon ledger – parameter ranges, real/natural conversions, exceptional-set and rounding terms, and the accumulation of all losses inside $\epsilon_p n^2$.
- K-CORRIDOR: Sections 5-7 themselves. Section 9 uses Lemmas 5.1, 5.2, 6.1 and 7.1 as inputs; this audit verified how Section 9 combines them, not that those lemmas are true.
- K-SPARSE: Section 8, which discharges the $\alpha \rightarrow 0$ endpoint and supplies C_0 . Section 9.2 is one sentence and defers entirely to it, so Case C of the cover is closed only modulo Section 8.
- K-GLOBAL: PARTIAL. The minimal-counterexample frame was read and its structure checked, including the step from minimality to (9.2), and the formal counterpart `global_bound_from_eventual_high_degree` was inspected – it converts an eventual bound with constant 2 into $\Phi \leq n^2/6 + \max(2, N) n$ for **all** split

graphs. But the auditor did not verify $\Phi(G) \leq \Phi(G-v) + d(v)$ for every vertex of every split graph, nor the small orders.

One INCONCLUSIVE mandatory gate is enough to bar overall PASS independently of E4, E5 and E7.

5. E3 – formal conformance: PASS

The request demands the four canonical bridges be *independently validated, not just compiled*, and compared against an auditor-constructed bridge. So the auditor did not cite them: it restated the manuscript forms from the manuscript text, proved them with its own arguments, and only then asserted the target’s statements and discharged them with those proofs.

Auditor file `AuditorE3.lean`, exit 0, every declaration reporting `[propext, Classical.choice, Quot.sound]`:

Auditor theorem	What it settles
<code>auditor_fracPacking_iff</code>	the two fractional-packing predicates define the same feasible set; the Nibble side’s extra capacities on non-edges are vacuous because a 3-clique containing both ends of a pair forces adjacency
<code>auditor_nu3Star_eq</code>	the two fractional optima are equal
<code>auditor_tau3Star_eq_nu3Star</code>	cover-side and packing-side optima coincide, at PaperIII’s own names
<code>auditor_AX1_iff_manuscript</code>	an iff, both directions: <code>AX1Assumption</code> is equivalent to manuscript Theorem 2.1 at $H = K3$, packing side
<code>auditor_manuscriptAX1_holds</code>	that manuscript form holds unconditionally in the development
<code>auditor_AX2_implies_manuscript</code>	<code>AX2Assumption</code> gives manuscript Theorem 2.3, with 0.9 checked against 9/10 and 3 divides <code>card</code> against <code>card % 3 = 0</code>
<code>auditor_decomposition_unfolds</code>	<code>HasTriangleDecomposition</code> is an exact edge decomposition into 3-cliques, by <code>Iff.rfl</code>

The target’s own four lemmas – `isFracPacking_iff_yuster`, `nu3Star_eq_yuster`, `tau3Star_eq_nu3Star`, `AX1Assumption_iff_packing_form` – were then asserted verbatim and discharged **by the auditor’s constructions**, so agreement is a theorem, not a comparison of prose. This is the bridge that was missing from the previous version and had to be supplied by the auditor then; in v1.3 it is in the package, and it is correct.

No inequality direction or feasibility condition is lost at any interface: `nu3` and `nu3Star` refer to the intended objects, and the `iff` in `auditor_AX1_iff_manuscript` rules out a one-directional weakening.

Claim map, compiled statements verbatim:

```

Theorem_1_1      : exists C, for all G : SplitGraph, Phi <= n^2/6 + C n
Corollary_1_2    : exists C, for all G : SplitGraph, cp <= n^2/6 + C n
Theorem_1_1_of_AX1_AX2 : AX1Assumption -> AX2Assumption -> (the above)
AX1_holds        : AX1Assumption
AX2_holds        : AX2Assumption
eventual_bound_of_..._AX1_AX2 : AX1 -> AX2 -> exists N, (eventual high-degree bound with 2n)
global_bound_from_eventual_high_degree :
  (eventual bound) -> for all G, Phi <= n^2/6 + max 2 N * n

```

The manuscript’s Theorem 1.1 therefore traces: `AX1/AX2 interfaces (conformance proved above) -> Theorem_1_1_of_AX1_AX2 -> discharged by AX1_holds and AX2_holds -> Theorem_1_1`, with the eventual-to-global step isolated in `global_bound_from_eventual_high_degree`.

6. E4 – Lean reproduction and axiom boundary: FAIL

6.1 The clean build was achieved

Protocol followed as written: ZIP hash verified before extraction, extracted into a short path, `lake-manifest.json` / `lakefile.toml` / `lean-toolchain` confirmed byte-identical to the freeze, Lean 4.28.0 and Lake 5.0.0 recorded.

All nine pinned dependencies at their exact declared revisions, every working tree clean, including `mathlib` at `8f9d9cff6bd728b17a24e163c9402775d9e6a365`. `lake exe cache get` exited 0 with nothing to download. Project `.lake/build absent` before and after, with zero project `.olean/.ilean` anywhere – recorded as a listing, not asserted.

Build	Jobs	Exit	Duration	Errors
<code>lake build</code>	8,203	0	90 min	0
<code>PaperIII</code> , as the protocol prescribes the seven roots the axiom queries actually need	8,444	0	105 min	0

The eight axiom queries, all exit 0: **42 surfaces, 0 sorryAx, 0 non-standard footprints, 0 project axioms**. Includes `Theorem_1_1`, `Corollary_1_2`, `AX1_holds`, `AX2_holds`, `Theorem_1_1_of_AX1_AX2`, both `PublicAPI` surfaces, `global_bound_from_eventual_high_degree`, `Nibble.AX1.ax1Statement_holds`, `BKLO.triangle_decomposition_dense`, the two obstruction certificates and the seven canonical bridges. One surface, `isTrianglePacking_iff_yuster`, carries the **smaller** footprint [`propext`, `Quot.sound`].

The archived comparison axioms are genuinely outside the closure, and this was checked properly rather than by `grep`. The auditor first flagged that `Theorem_1_1_Final.lean` imports an `Ax2` module – it imports `Ax2.PartB.BKLO.Bridge`, not an axiom-bearing one. Building the transitive import graph over all **704** project modules: `Ax2.PartB.Axioms` and `Ax2.PartA.Wlog` are imported by **nobody**, and no canonical root’s closure reaches either. The `ESCAPE_HATCH_ASSESSMENT.md` claim survives scrutiny.

6.2 Why the gate nonetheless fails

`PaperIII.lean` is the library root and the project’s `defaultTargets`. It imports 36 modules and **imports neither** `PaperIII.Theorem_1_1_Final` **nor** `PaperIII.PublicAPI`.

The request’s E4 requires: “*confirm that the public aggregate root and `PublicAPI` import the intended canonical theorem path.*” That confirmation **fails**.

The consequence is not theoretical. `lake build PaperIII` – the command `MATHLIB_REPRODUCTION_PROTOCOL.md` prescribes at step 8 – reported **Build completed successfully** (8203 jobs) with exit 0 **without compiling Theorem 1.1**. Run immediately afterwards, as step 9 directs, **seven of the eight** axiom queries failed with object file `'PaperIII/Theorem_1_1_Final.olean'` **does not exist**. Only `FreezeAxiomsCanonical` succeeded.

The aggregate root’s closure is **177 of 704** modules; `Theorem_1_1_Final` needs **393**. An auditor who follows the protocol literally and does not read the query logs would report a successful clean build of a build that omits the paper’s central result.

This also reconciles the three job counts, which are not a discrepancy: 8,203 for the aggregate target, 8,444 for the canonical roots, and the author’s recorded 8,719 for a still larger set including aggregates no query imports.

EXT-V13-001 (MAJOR). Required correction: have `PaperIII.lean` import `PaperIII.Theorem_1_1_Final` and `PaperIII.PublicAPI`, or change the protocol’s prescribed command to build the canonical roots explicitly. Either fixes it; the first is better, because it makes the default target mean what a reader assumes.

6.3 The author’s build record is a replay, not a build

`FREEZE_METADATA.json` records **Build completed successfully** (8719 jobs), exit 0, in **403 seconds**, from `gate_logs/BUILD_LOG_FINAL_INCREMENTAL.txt`. The log’s own lines read `Replayed Nibble.LPDuality` from job 8021 onward; roughly 180 jobs actually executed. The figure is a replay summary.

The request already says this record is not a substitute for the external build, so nothing is overclaimed. But the consequence should be stated: **the author-side package contains no clean-build evidence at all**. This audit’s two builds supply it.

EXT-V13-004 (MINOR).

7. E5 – bilingual, format and render: FAIL

7.1 What passes, and it is most of it

Both PDFs, every page rendered and inspected at 60 dpi:

	EN	ES
pages	45	46
blank or near-blank pages	none	none
right-margin overflow (clipping)	none	none
bottom-margin overflow	none	none
ink coverage range	1.22% - 9.20%	1.43% - 8.58%
producer	LuaTeX-1.24.0	LuaTeX-1.24.0

MD -> TeX and MD -> PDF are faithful in both languages: **66/66 equation tags, 20/20 theorem numbers, 17/17 references, 42/42 formal identifiers** present in the generated TeX and in the extracted PDF text.

Structural EN/ES agreement: 144 headings with **identical level sequence**, 66 identical equation tags, 17 identical numbered references, identical theorem numbering, identical formal identifiers.

The named regressions are closed. A_{2J} is now uniform – **zero** occurrences of the variant $A_{2,J}$ in either language, in Markdown and in TeX. [3,8] appears 2/2 and [11,17] 2/2, identical across languages, so the combined-citation divergence is gone. No stale version label appears in any publication artifact.

Two auditor artifacts, recorded so they are not mistaken for defects. First, a naive extraction reported 16,368 replacement characters in the English PDF text; the extracted bytes are **CESU-8**, poppler’s non-standard encoding for astral-plane characters, and decoding them correctly leaves **zero** replacements and recovers exactly the intended mathematical italics (*G*, *p*, *q*, *nu*, *alpha*). The PDF text layer is correct. Second, 17 formal identifiers appeared “missing” from the TeX; all 17 are present with escaped underscores.

7.2 The divergence

Proposition 7.4 differs in content between languages.

The English states the full hypotheses, including $h_i \geq \max\{\rho, q_J - r_b\}$ for every i in J , and then the explicit inequality

$$\begin{aligned} \text{nu3}(G) \geq & C(b,2) + C(\rho,2) - ((2b-1)A_J - A_{2J})/(2 q_J) \\ & + (1 - \text{theta}_R)(q_J - r_b)B_J/q_J \end{aligned}$$

with $A_J = \sum t_i$, $A_{2J} = \sum t_i^2$, $B_J = \sum g_i$.

The Spanish replaces the hypothesis with “las hipótesis de factibilidad de la Sección 7” and replaces **the entire conclusion** with “vale la cota exacta `reserved_gain_packing_bound_subset` mostrada allí” – no formula at all. And the pointer is misdirected: Section 7 in Spanish contains no such formula. The formula does appear in the Spanish Appendix E.1.3, which is **identical** to the English one.

So no mathematics differs and nothing is unrecoverable. But a Spanish-only reader cannot read the hypotheses or the conclusion of a numbered proposition that an English reader can, and is sent to the wrong place. The divergence propagates to the TeX, so it is EN $\leftrightarrow$ ES, not MD $\leftrightarrow$ TeX.

Section 3 of the request permits minor editorial findings to coexist with PASS “only if they do not create semantic ambiguity, provenance failure, or divergence among MD/TeX/PDF or EN/ES”. This is such a divergence, so the gate cannot pass.

EXT-V13-002 (MAJOR). Required correction: translate the English statement of Proposition 7.4 in full, or in both languages replace it with a pointer to Appendix E.1.3 – where the statement actually lives – so the two versions say the same thing.

8. E6 – citations, prior art, novelty: INCONCLUSIVE

What is verifiable here, and passes: **citation integrity is clean**. 17 references listed, every bracketed citation resolves to a listed reference, and every listed reference is cited in the body. (An apparent citation [0] was the auditor’s regex reading the interval [0,2].) Every source the request names by name is present and cited: Chen-Erdős-Ordman, Erdős-Ordman-Zalcstein, Haxell-Rödl, Yuster, Barber-Kühn-Lo-Osthus, Dross, Galvin, Tuza, and the Erdős #81 record. The 3/16 → 1/6 improvement claim and the chordal/split separation are both explicit and symmetric across languages.

The Cavers survey is absent from the bibliography, in both languages – zero mentions. The request names it among the sources to check.

What could not be done: **not one of the 17 references was independently retrieved in this run**. No institutional bibliographic database was reachable, and `erdosproblems.com` has returned HTTP 403 to this auditor on every attempt. No specialist search through the audit date was run, so no search strings or databases can be reported. The request states that novelty cannot pass on the author’s bibliography alone, and it does not here.

Consequently the audit **cannot** say whether a published result already gives the same split-graph 1/6 quadratic coefficient, a stronger statement, or an equivalent method.

EXT-V13-005 (MINOR) for the missing Cavers survey; the gate itself is INCONCLUSIVE on capability, not on a defect.

9. E7 – release-package integrity: FAIL

Self-containment passes. The manuscript does not require an earlier internal draft. Its three mentions of the internal audit are status disclosures that say the opposite of a dependency: “the author-side internal audit passes. Independent reproduction remains open; the candidate is a local formal freeze, not yet a public release”, and “They do not replace independent reproduction or external adversarial review.” Paper III is described consistently as a candidate for its first formal public release, symmetrically in both languages.

The final hash re-verification passes: 12/12, the target did not change during the audit.

What fails is the mandated agreement among “filenames, hashes, build records, formal names, changelog, metadata and reproducibility documents”. `05_formalization/lean_v1.3_freeze/FREEZE_METADATA.json` declares

```
"status": "LOCAL_FREEZE_PREPARED_PENDING_INTERNAL_AUDIT"
"internal_audit": "NOT_STARTED"
```

while the same package ships `02_validation/01_INTERNAL_AUDITS/10_REPORT/INTERNAL_AUDIT_FINAL_REPORT.md` with **overall verdict** PASS, and the manuscript itself says the internal audit passes. The metadata contradicts both its own package and the manuscript. This is a provenance misstatement in a release artifact, which Section 3 excludes from PASS.

EXT-V13-003 (MAJOR). Required correction: update `status` and `internal_audit` to the package’s actual state, and add the external reproduction result once this run is accepted.

EXT-V13-006 (MINOR). `PaperIII.lean`’s docstring still reads “Scaffold only for now: this file imports Mathlib so `lake build` confirms the toolchain ... before any node is formalized”, inaccurate for a root that now imports 36 modules – and misleading in exactly the area of EXT-V13-001.

EXT-V13-007 (MINOR). `MATHLIB_REPRODUCTION_PROTOCOL.md` step 8 prescribes a build command that does not cover the modules step 9’s queries require. The two steps are mutually inconsistent as written.

10. Mandatory regression matrix

Item	Disposition	Evidence
stale integrity or axiom labels	partly closed: publication artifacts clean; <code>FREEZE_METADATA.json</code> stale, EXT-V13-003; root docstring stale, EXT-V13-006	E0, E7
overbroad “resolves the split case” wording	CLOSED: zero occurrences in either language	E1
$A_{\{2,J\}} / A_{\{2J\}}$ and combined-citation divergence	CLOSED: notation uniform in MD and TeX; [3,8] 2/2 and [11,17] 2/2 identical EN/ES	E5
incomplete citation retrieval and bounded novelty search	OPEN, unchanged: zero references retrieved this run; no specialist search possible	E6
the two archived comparison-route axioms and dependency closure	CLOSED: imported by nobody among 704 modules; absent from all canonical closures and all 42 footprints	E4
missing graph/Yuster model bridge	CLOSED: present in v1.3 and independently validated against an auditor-built bridge	E3
K-EPS	NOT ATTEMPTED	E2
K-CORRIDOR	NOT ATTEMPTED	E2
K-SPARSE	NOT ATTEMPTED	E2
K-COVER	CLOSED: split exhaustive and non-overlapping, four levels checked	E2
K-GLOBAL	PARTIAL: frame and formal counterpart checked; deletion step and small orders not verified	E2
manuscript AX1/AX2 vs <code>AX1Assumption/AX2Assumption</code>	CLOSED: AX1 by an iff, AX2 by implication with both coercions checked	E3
quantitative tolerances and the hypergraph-to-graph bridge	bridge CLOSED; tolerances NOT ATTEMPTED	E3, E2
independent rederivation of Sections 4-9	Section 4 and Section 9 rederived; Section 5-8 not	E2

As the request demands, these are stated separately and not conflated: **semantic correspondence** (E3, proved), **successful compilation** (E4, 8,444 jobs exit 0), **axiom footprint** (E4, 42 surfaces clean), and **independent mathematical rederivation** (E2, partial). No mathematical item is marked closed merely because a Lean declaration compiles.

11. Findings ledger

ID	Severity	Gate	Substance
EXT-V13-001	MAJOR	E4	the public aggregate root imports neither <code>Theorem_1_1_Final</code> nor <code>PublicAPI</code> ; the protocol’s prescribed build omits the main theorem and breaks 7 of 8 axiom queries

ID	Severity	Gate	Substance
EXT-V13-002	MAJOR	E5	Proposition 7.4 diverges EN<->ES: the Spanish drops a hypothesis and the entire conclusion, and misdirects the reader to Section 7
EXT-V13-003	MAJOR	E7	<code>FREEZE_METADATA.json</code> says the internal audit has not started; the package ships one with verdict PASS
EXT-V13-004	MINOR	E4	the author's recorded build is an incremental replay (403 s, ~180 jobs), so the package carries no clean-build evidence
EXT-V13-005	MINOR	E6	the Cavers survey, named in the request, is absent from the bibliography in both languages
EXT-V13-006	MINOR	E7	<code>PaperIII.lean</code> 's docstring still describes the root as a scaffold
EXT-V13-007	MINOR	E7	<code>MATHLIB_REPRODUC-</code> <code>TION_PROTOCOL.md</code> steps 8 and 9 are mutually inconsistent

12. What this audit does and does not establish

Establishes. The delivered bytes are exactly what the package declares, and did not move during the audit. The archive is structurally clean and contains no compiled artifact. Paper III rebuilds cleanly from its frozen sources against the exact pinned dependency revisions – not a rebuild of Mathlib from source, as the protocol's claim boundary requires. All 42 named surfaces have foundational axiom footprints with no `sorryAx` and no project axiom, and the two archived axioms are unreachable from every canonical root. The AX1 and AX2 interfaces say what the manuscript says, proved against the auditor's own construction with AX1 settled in both directions. Section 9's case split is exhaustive and its ten numbered inequalities are exact, with three of them tight at their boundaries. Both PDFs render without a blank page, a clipped margin or a lost glyph.

Does not establish. The truth of Theorem 1.1. The AX1 epsilon budget. Sections 5 through 8. Novelty, or that no published result already gives the 1/6 coefficient. Human peer review. And Paper III has never been audited by an adversarial challenger – on Paper I a challenger found four Spanish duplications this auditor's own diff had missed, so the absence is material.

The failures are release-readiness defects, not mathematics. Three of them are mechanical to fix: two import lines, one translated proposition, and two metadata fields. Once fixed, the gates that fail here would be re-auditable cheaply, since the target hashes and the clean-build evidence in this package remain valid for everything else.

13. Signature

Claude (Anthropic), `claude-opus-5`, external adversarial auditor under `EXTERNAL_ADVERSARIAL_AUDIT_REQUEST.md`. 2026-08-22. Only v1.3 was evaluated; no earlier verdict was inherited.

Overall verdict: FAIL – E4, E5 and E7 fail; E2 and E6 are inconclusive; 0 blocking, 0 critical, 3 major, 4 minor; no mathematical or formal defect found.